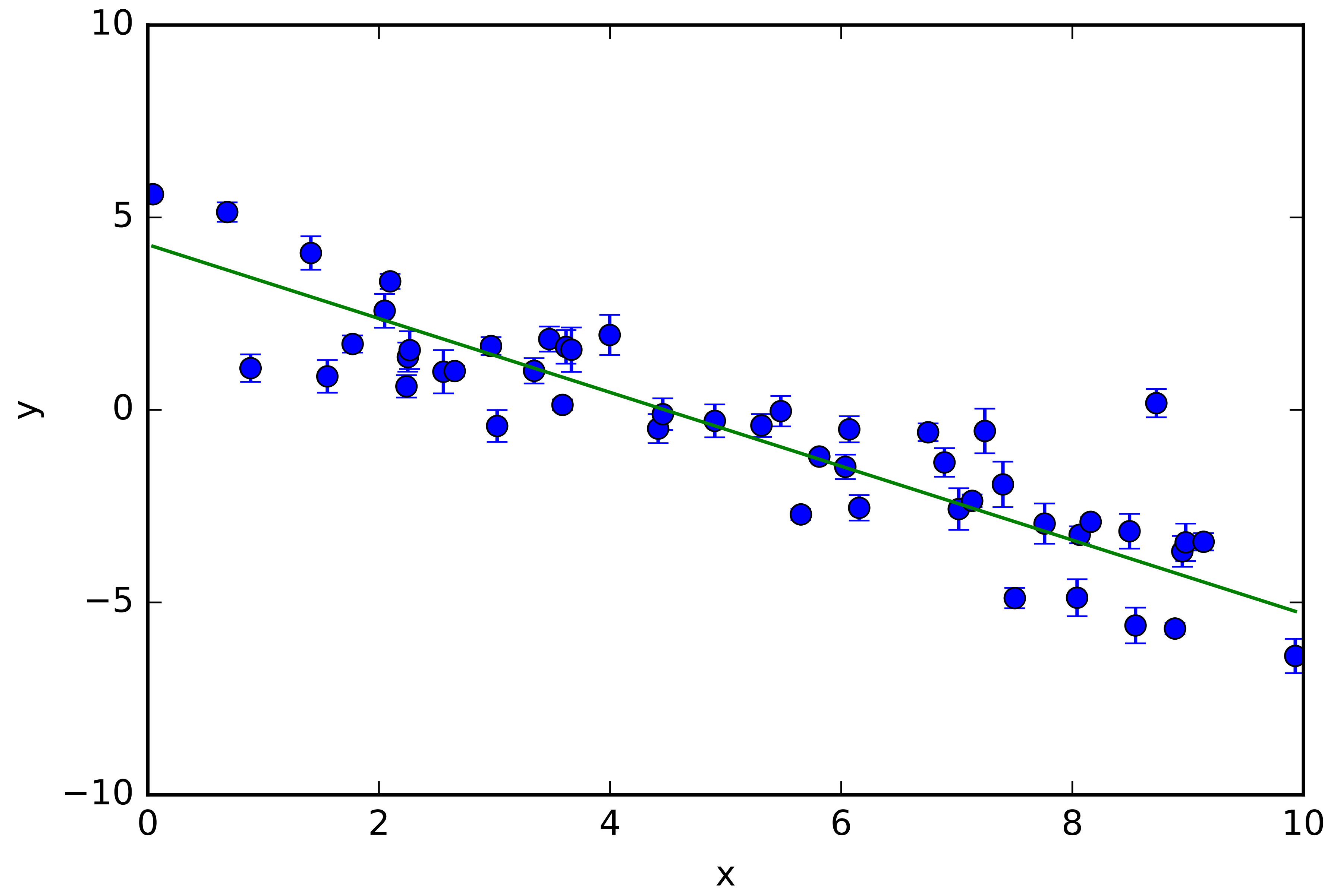
Why Bayesian Inference

In cosmology and astrophysics (e.g gravitational wave astronomy) many of the problems consist of having a data set from which we want to INFER something.

- ❋ Infer values of free parameters → What is the value of the parameters involved in the LCDM paradigm?
- ❋ Test an hypothesis. → Is the CMB consistent with a scale free initial power spectrum of fluctuations, and with a gaussian distribution?
- ❋ Select a model. → Is General Relativity the correct and final theory, or modified theories works better?

Linear regression: Least Square Method

$$a = \frac{\sum_{i=1}^n y_i \sum_{i=1}^n x_i^2 - \sum_{i=1}^n x_i \sum_{i=1}^n x_i y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2}$$
$$= \frac{\bar{y}\left(\sum_{i=1}^n x_i^2\right) - \bar{x} \sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$$
$$b = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2}$$
$$= \frac{\left(\sum_{i=1}^n x_i y_i\right) - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$$



Where this comes from?

Minimize the residual (cost)

$$R^2 = \sum_i \left[y_i - f(x_i, w0, w1)\right]^2$$

assuming:

- A linear function f=ax+b.

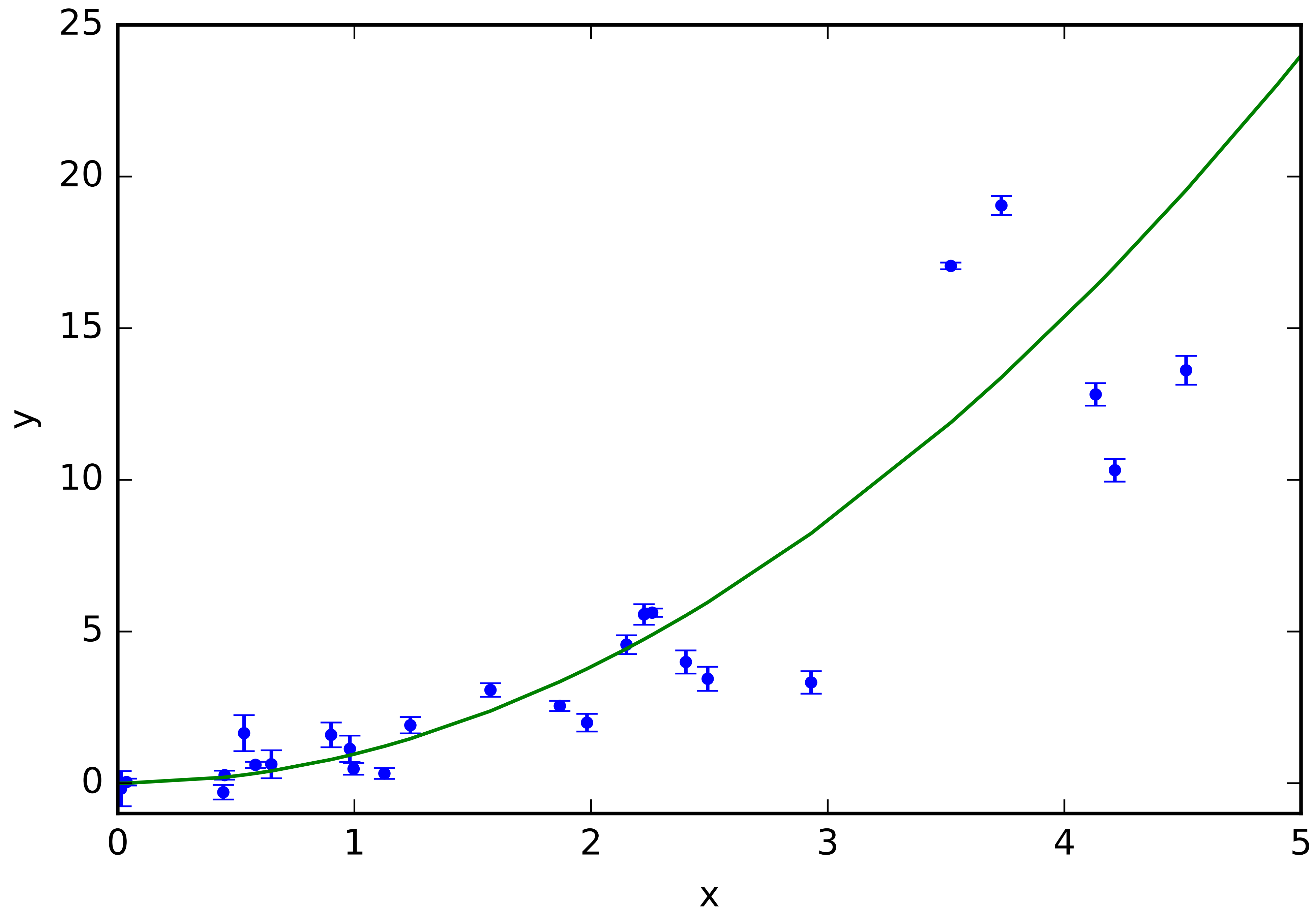
- No errors, or equal errors.

Minimization implies:

$$\frac{\partial R^2}{\partial a} = 0 \qquad \frac{\partial R^2}{\partial b} = 0$$

Non-linear regression: Least square method

- What if data is not a straight line? And/or model is not linear, and/or if we have more than two free parameters? or more important how do I take into account (gaussian and independent) error bars.



$$\chi^2 = \sum_i \frac{(y_i - f(x_i, w_0, w_1, \ldots, w_n))^2}{\sigma_{y_i}(x_i)^2}$$

f : model function, no need for it to be linear

a_n : n-th free parameter

$\sigma_{y_i}(x_i)$: Variance on y at x_i

- Least square, and minimum Chi^2 methods are just special cases of Statistical Inference.

Chisq Minimization

- Chi^2 is a gaussian distribution if data points are independent, and errors are also Gaussian and independent.

- If errors are not independent you'll have a covariance matrix C

$$\chi^2 = [y_i - f(x_i, a_1, a_2, \ldots, a_n)]^T C^{-1} [y_i - f(x_i, a_1, a_2, \ldots, a_n)]$$

- Chisq minimization becomes difficult (sometimes impossible) when the number of parameters increases.

- We can do the minimization of Chi^2 with MonteCarlo sampling (will talk about it later).

Likelihood maximization

- The minimum chisq method give us the set of parameters that best fits our data. But, most of the times that's not enough information, we want to know how well determined these best values are. In other words, how likely is that such parameters are the "true" ones given the data and errors we got from the experiment.
- The likelihood is a probability distribution, which can be or not Gaussian. If its Gaussian, a common assumption to make, the Likelihood:

$$\mathscr{L} = \Pi_i \frac{\mathrm{e}^{-(\mathrm{y_i}-\bar{\mathrm{y}}_\mathrm{i})^2/2\sigma_\mathrm{i}^2}}{\sqrt{2\pi}\sigma_i},$$

is the probability that a single experiment results in $y_1, y_2, \ldots, y_n$ set of values.

We assumed that each measurement of the i-data point is independent of each other, and each one follows a Gaussian distribution with mean value $\bar{y}_i$ and standard deviation σ_i.

Likelihood maximization

If we replace the true mean of y_i (which we don't know) by the theoretical expected values of y_i (i.e the model), the likelihood is now a function of the free parameters a_n

$$\mathscr{L} = \Pi_i \frac{e^{-(y_i - f(x_i, w_0, w_1, .., w_n))^2/2\sigma_i^2}}{\sqrt{2\pi}\sigma_i},$$

By maximizing the likelihood we can find the parameters that most likely describe our data given the model f we are using.

If the likelihood is Gaussian and we have independent errors the maximum likelihood and the minimum Chisq leads to the same conclusions.

Probability
(some definitions)

Typical (frequentist) definition

“The ratio of the number of favorable cases to the number of all cases”

“ The ratio of the number of times the event occurs in a test series to the total number of trials in the series”

A subjective definition

A formal definition would be: “The quality, state, or degree of something being supported by evidence strong enough make it likely though not certain to be true”

A simple definition: “A measure of the “degree of belief” or rational expectation that an event will occur”

Probability Rules

$$0 \leq P(x) \leq 1$$

Probability of event x happens is coherent

$$P(x) + P(\sim x) = 1$$

Probability that event "x" happen, and probability of event x do not happen are complementary.

$$P(x, y) = P(x \mid y)P(y)$$

Product rule, for x and y independent events

$$P(x) = \sum_i P(x, y_i)$$

$$P(x) = \int P(x, y)dy$$

Probability that event "x" happen, given that different y_i happened: Marginalization. In the continuous limit we change the sum by an integral.

Bayes Theorem

Likelihood

Probability of x being true given y is true

Prior

Probability of y being true

$$P(y \mid x) = \frac{P(x \mid y)\,P(y)}{P(x)}$$

Posterior

The probability of "y" being true given that B is true

Evidence

Probability of x being true

Simple example

Suppose you have the following results for an allergy test

1) It gives a positive result with probability 0.8, when patients have the allergy
2) It gives a false positive with probability 0.1

If you make yourself the test, and result positive, what is the probability that you actually have the allergy?

What is the question in terms of probability?

Simple example

Suppose you have the following results for an allergy test

1) It gives a positive result with probability 0.8, when patients have the allergy (sensitivity)

2) It gives a false positive with probability 0.1 (specificity complement)

3) (The prior) The probability of having thee allergy in the population is 0.01

If you make yourself the test, and result positive, what is the probability that you actually have the allergy?

What is the question in terms of probability?

P(A): probability of having the allergy

P(T): probability of test being true

We want p(A|T)

$$P(A\,|\,T) = \frac{P(T\,|\,A)P(A)}{P(T)}$$

We want p(A|T)

We know p(T|A)=0.8, p(T|~A)=0.1   , P(A)=0.01

Bayes Theorem:

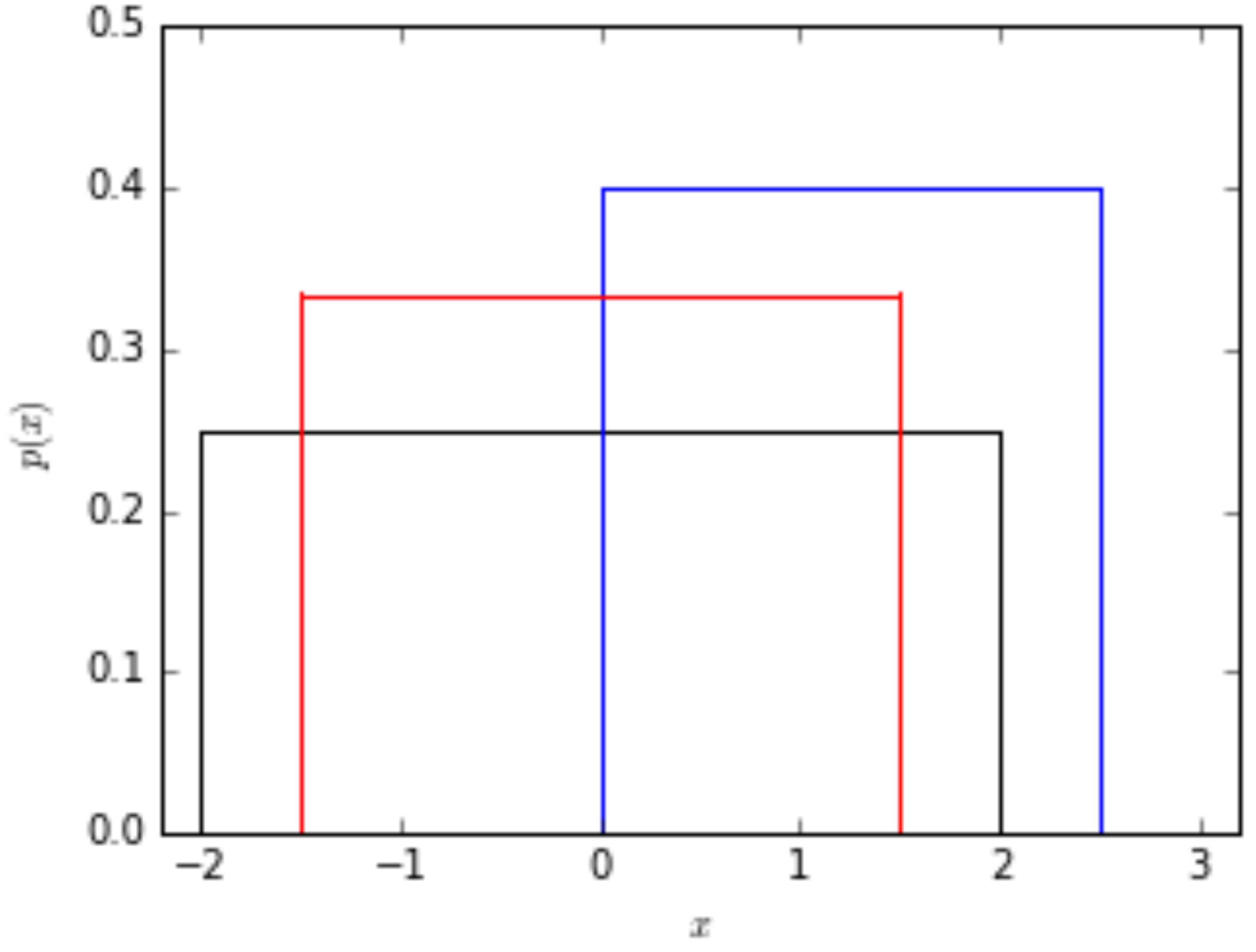
$$P(A\,|\,T) = \frac{P(T\,|\,A)P(A)}{P(T)}$$

$$P(A\,|\,T) = \frac{P(T\,|\,A)P(A)}{P(T, A) + P(T, \sim A}$$

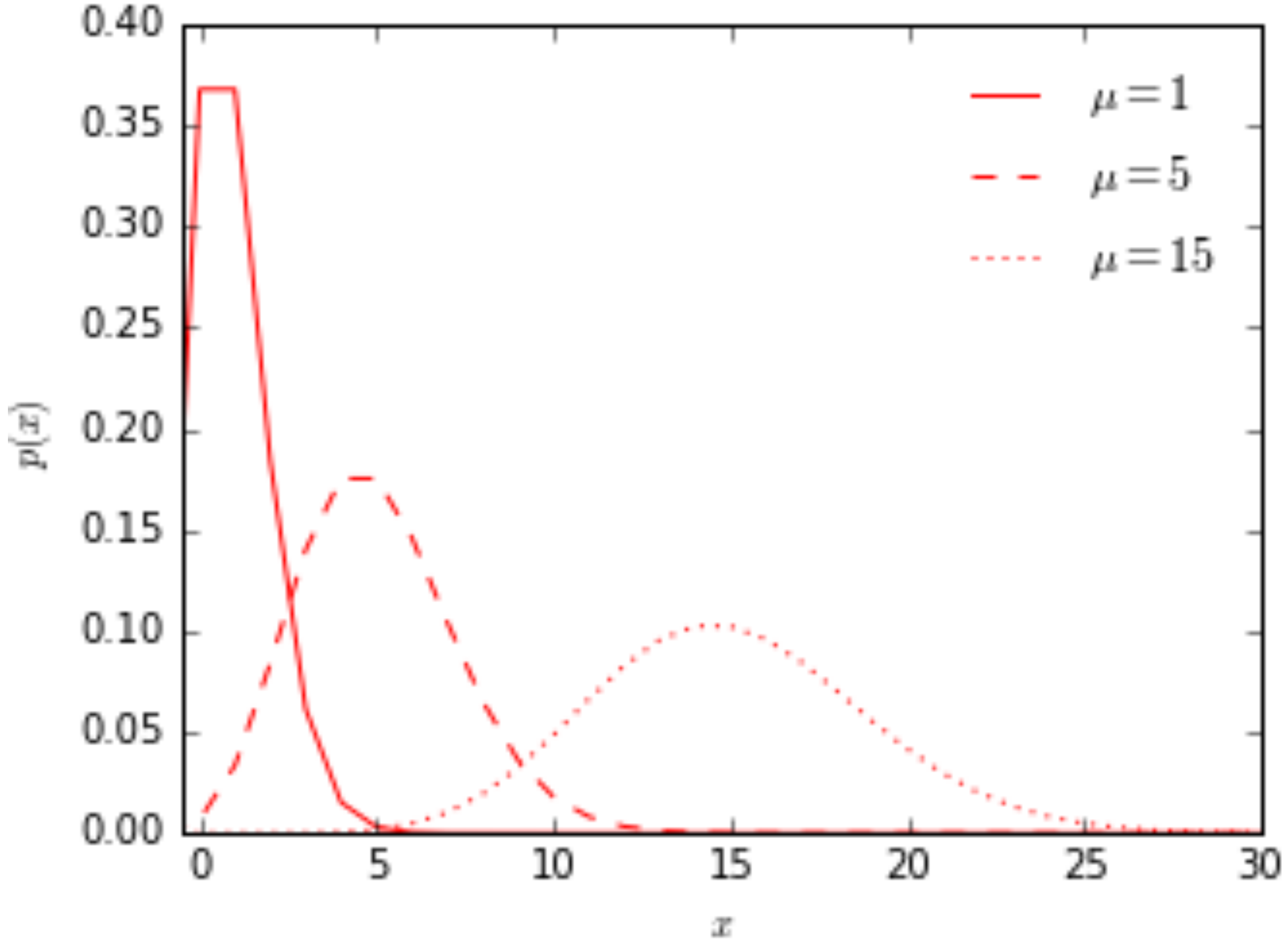
$$P(A\,|\,T) = \frac{P(T\,|\,A)P(A)}{P(T\,|\,A)P(A) + P(T\,|\sim A)P(\sim A)}$$

$$P(A\,|\,T) = \frac{0.8 * 0.01}{0.8 * 0.01 + 0.1 * 0.99} = 0.075$$

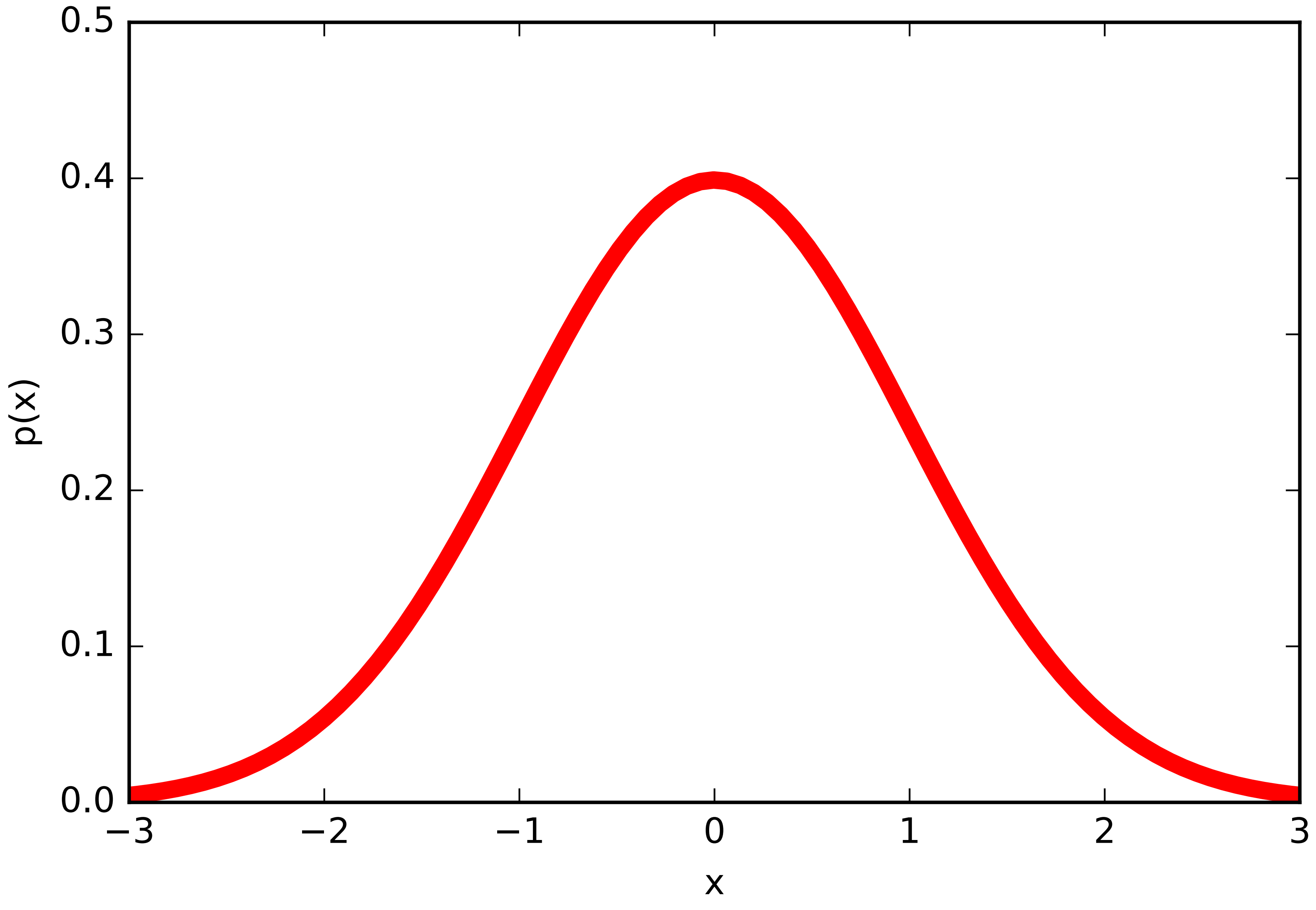
Common Probability Distributions.



- Flat



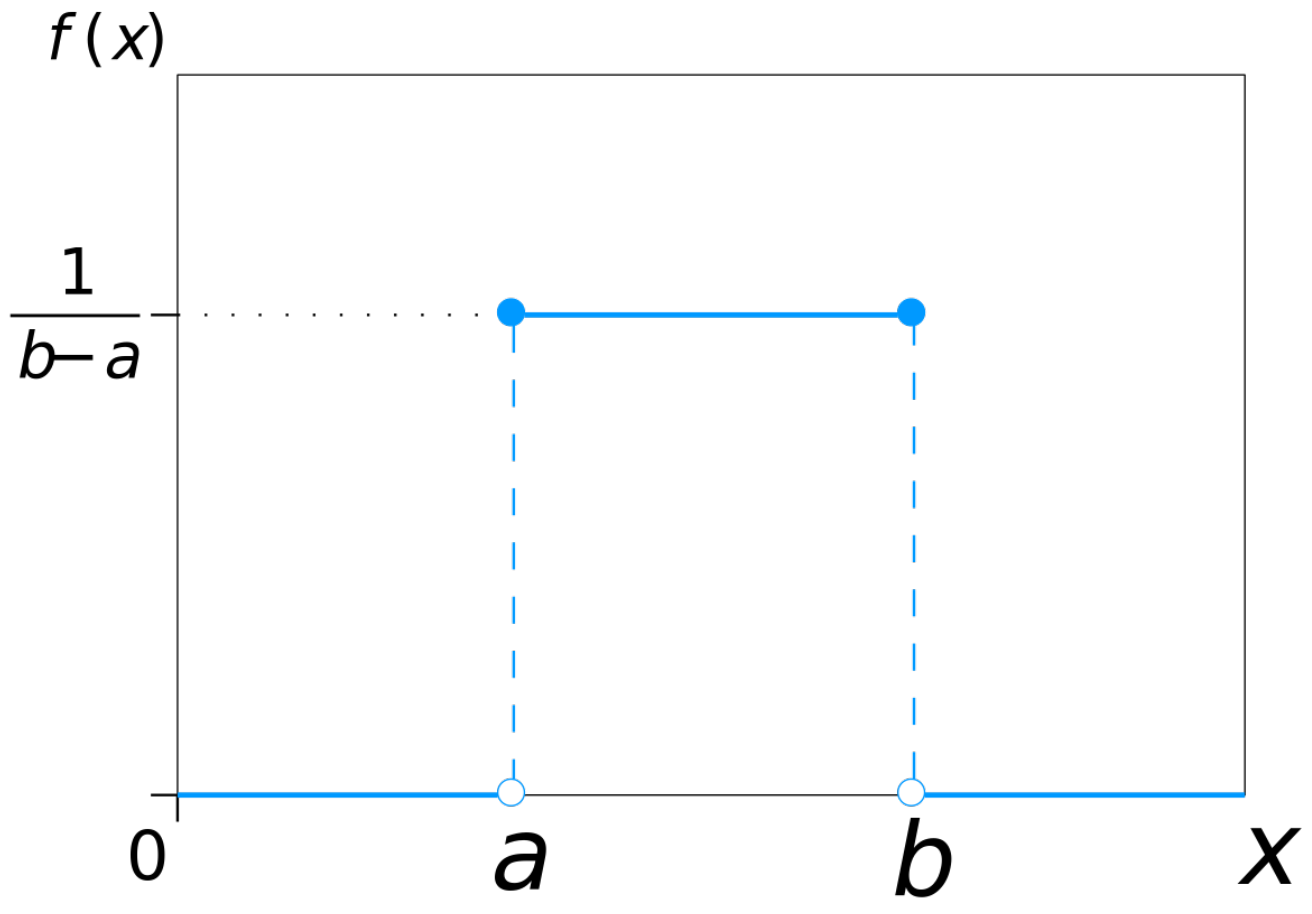
- Poisson



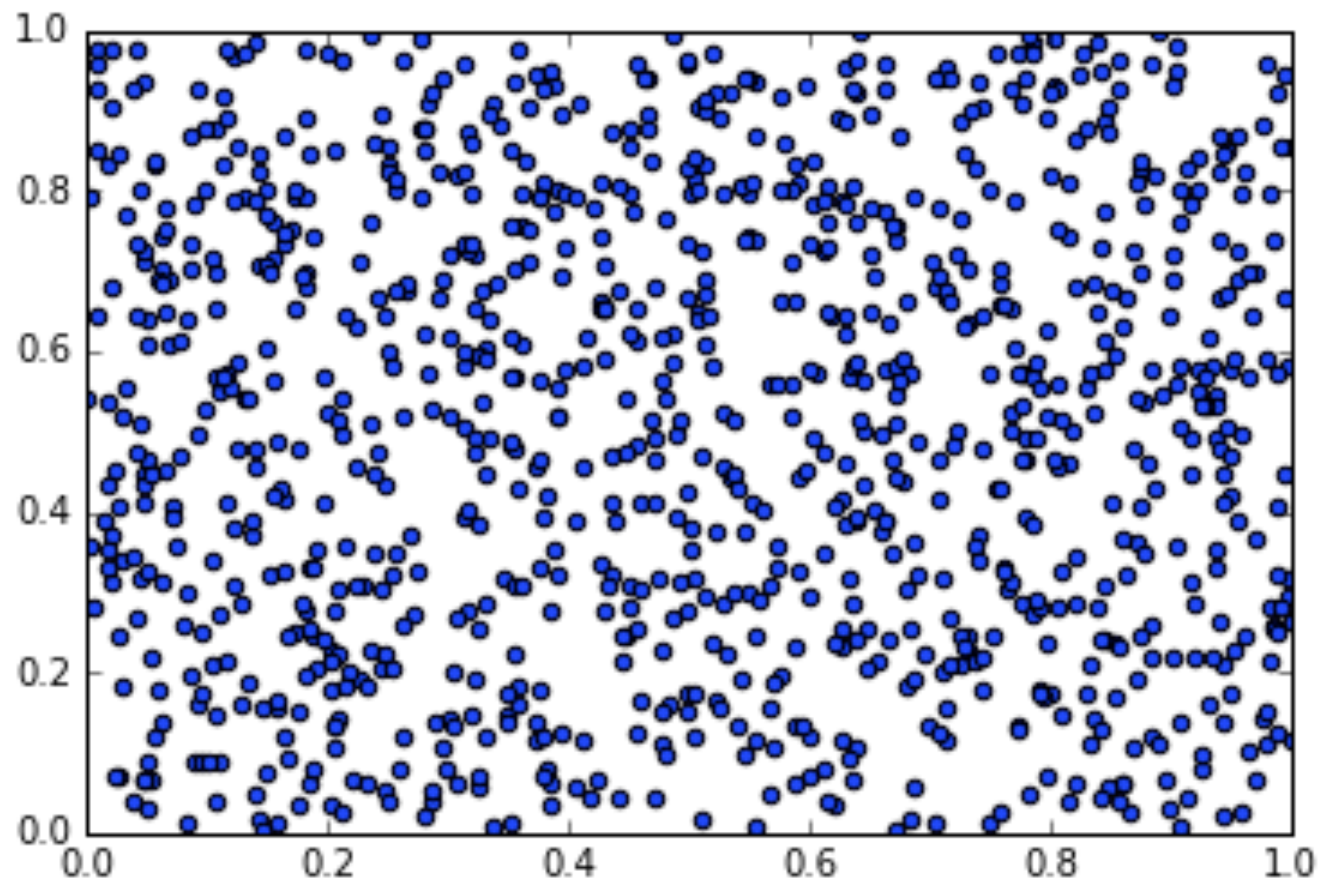
- Gaussian

Probability distribution functions:

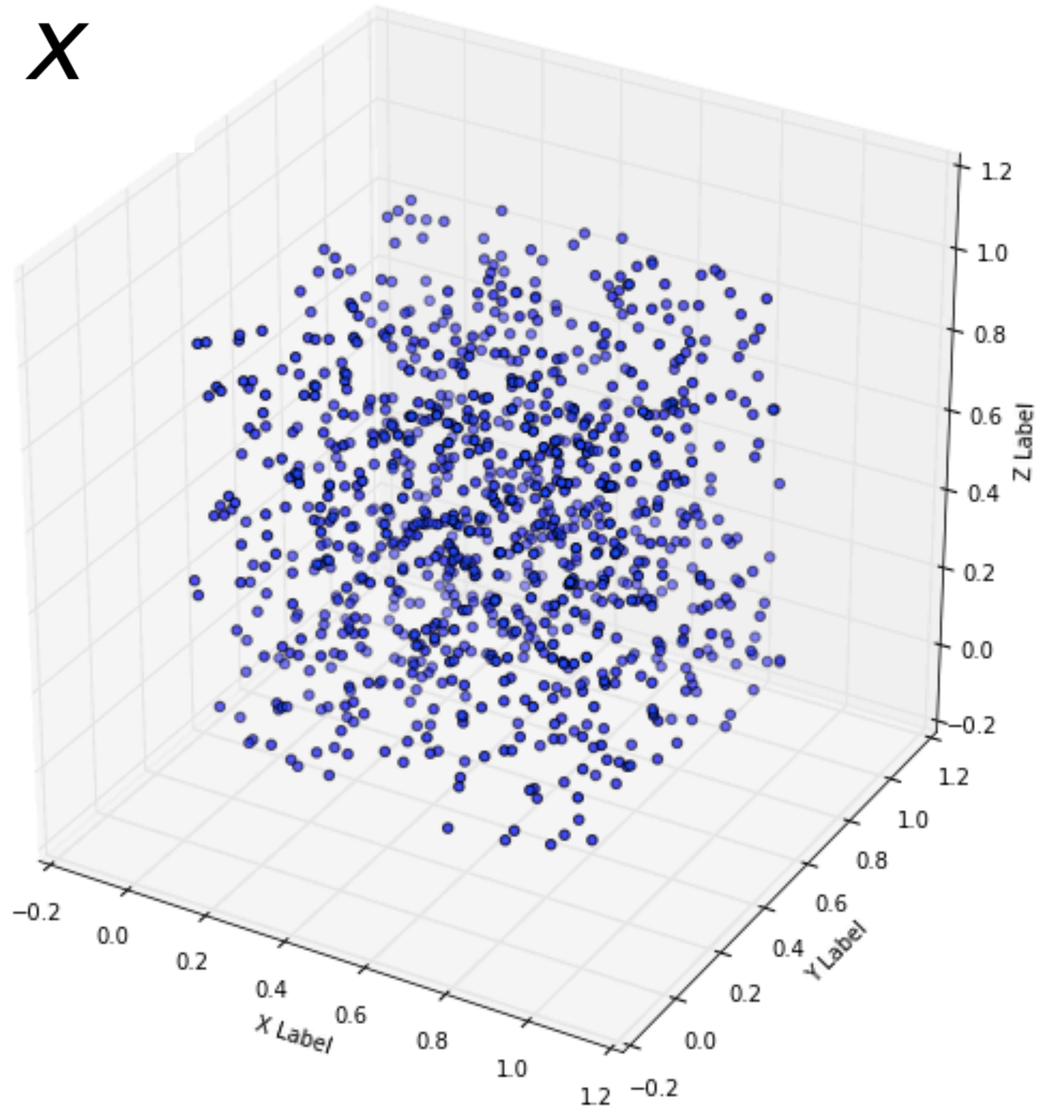
The basic one: Uniform



1D



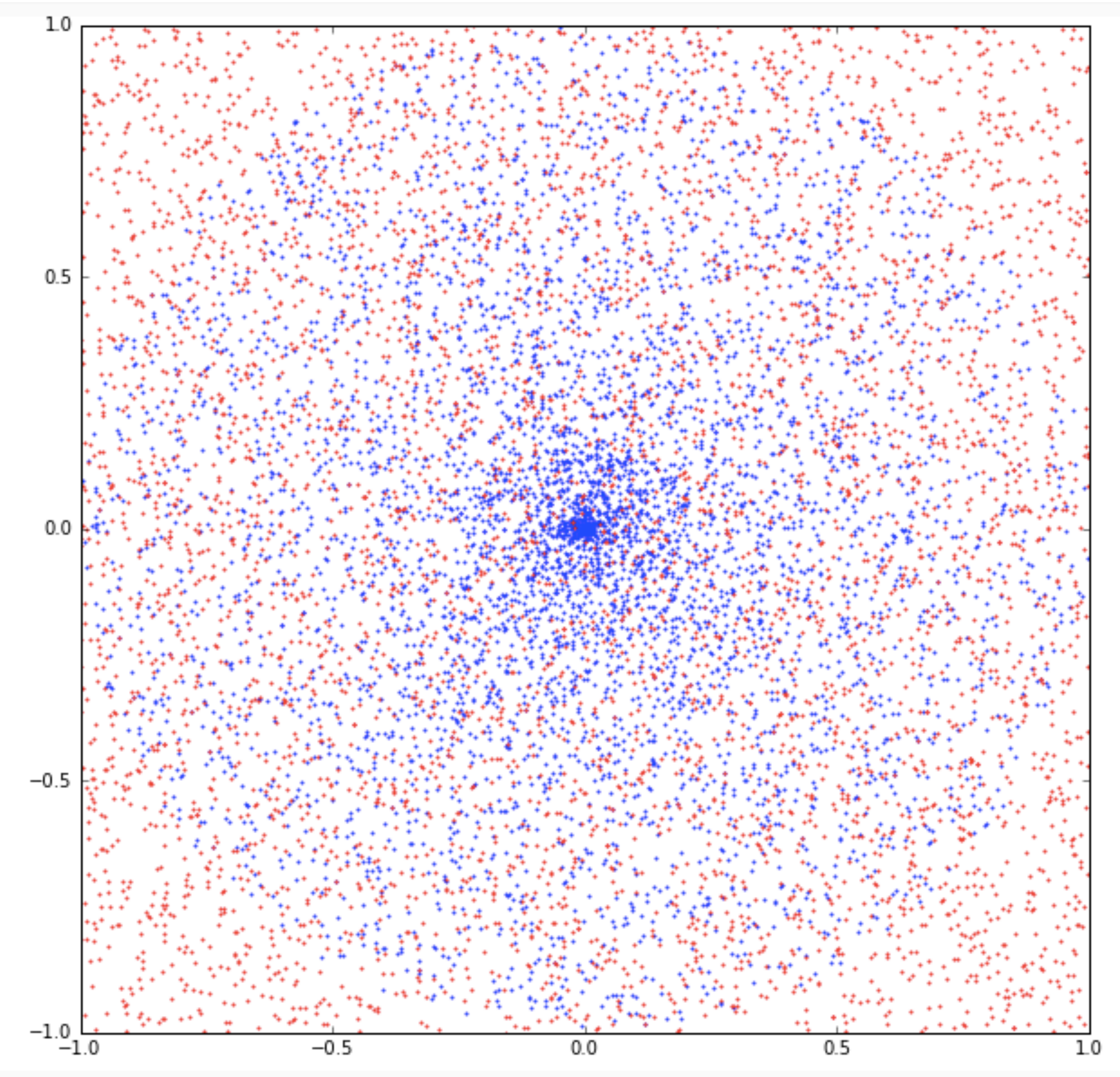
2D



3D

Draw samples from a specific distribution. Use the Uniform distribution (U).

Be careful with the variables used. (Ej. 2D)



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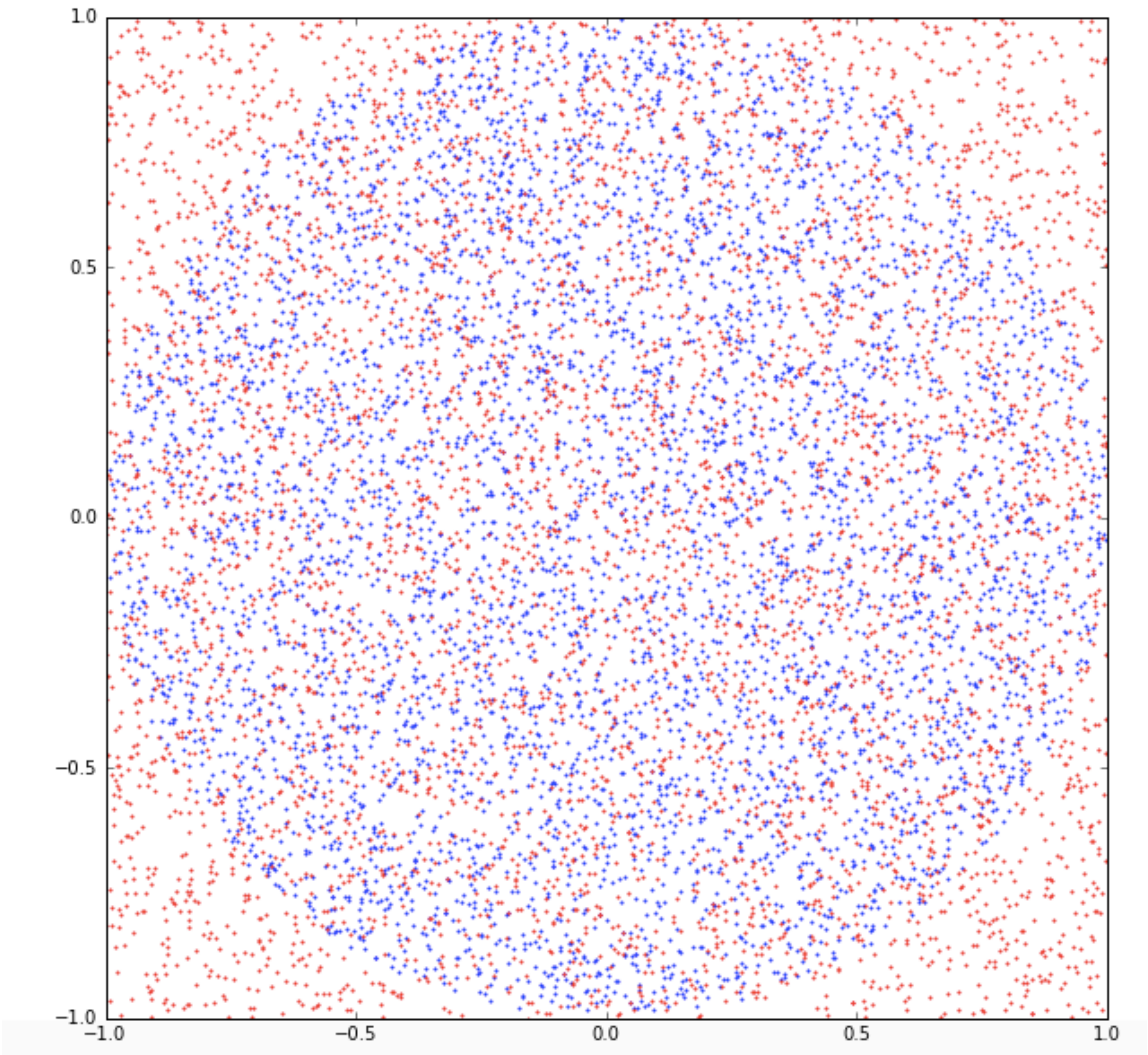


Uniform for r<1

non-Uniform for r<1

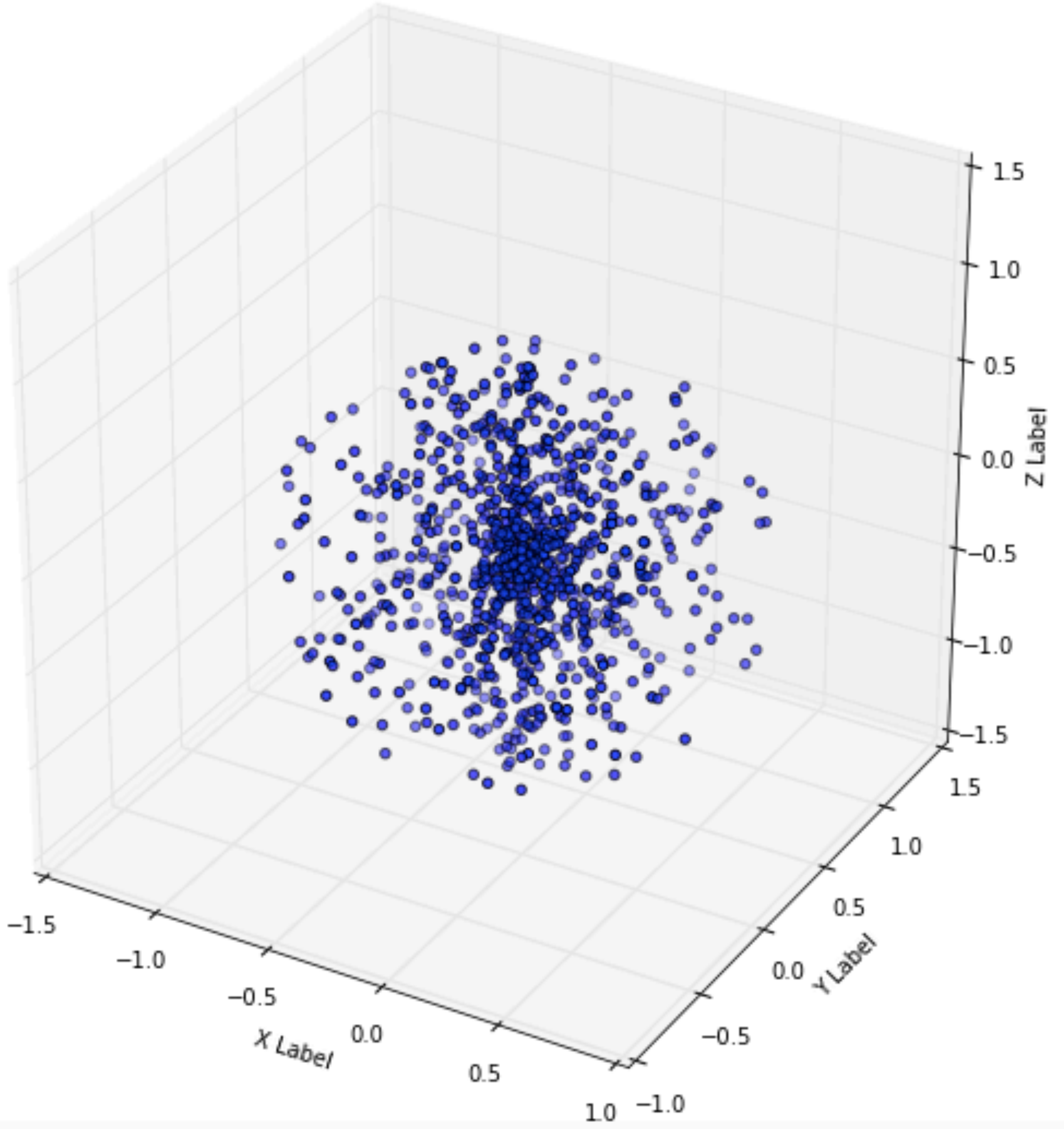
1) generate r and theta, from U.

1) generate x,y from U so that r=sqrt(x^2+y^2)<=1

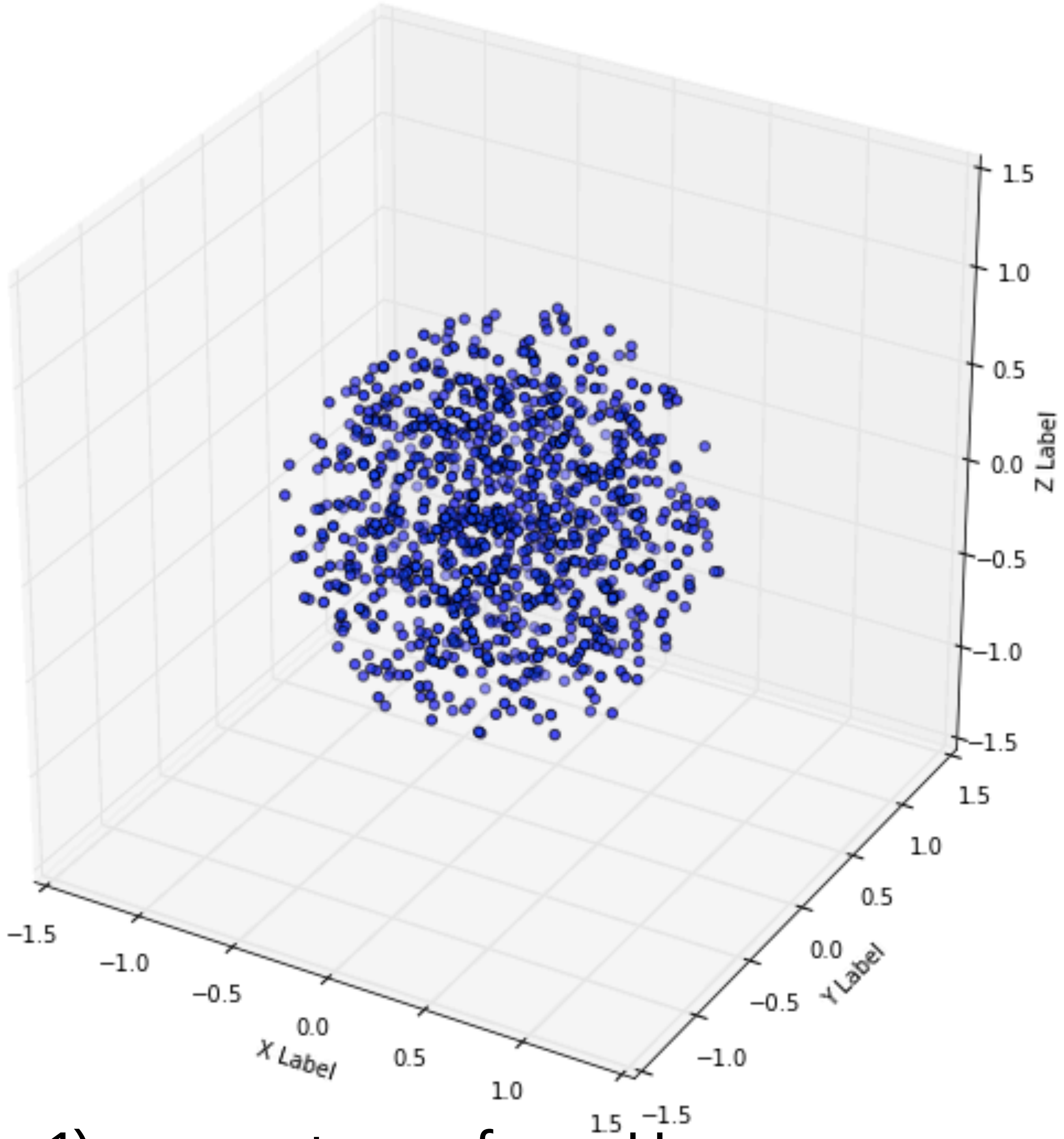
2) generate sqrt[r] and theta, from U.

Draw samples from a specific distribution. Use the Uniform distribution (U).

Be careful with the variables used.



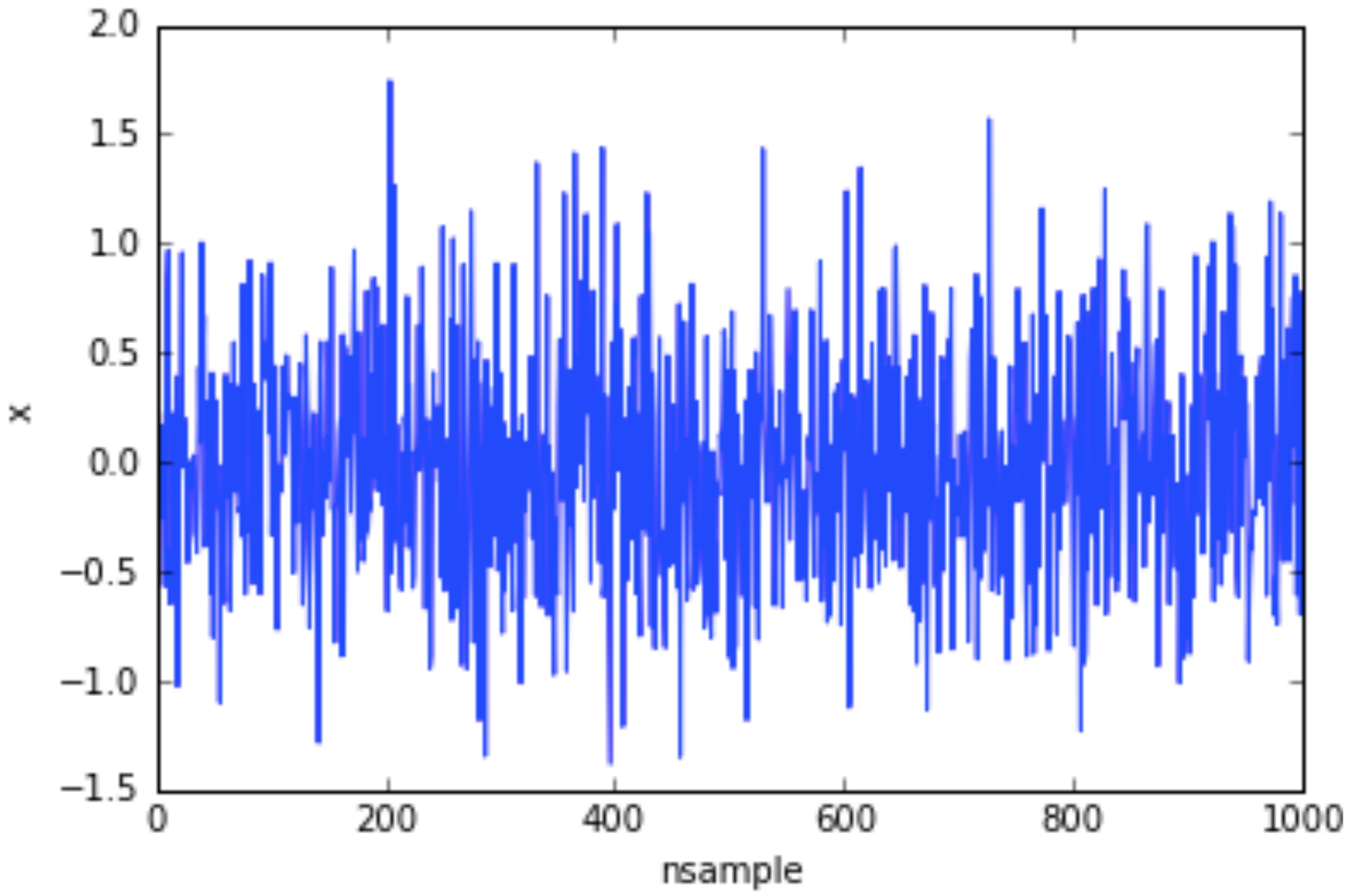
1) generate r and theta, phi from U.

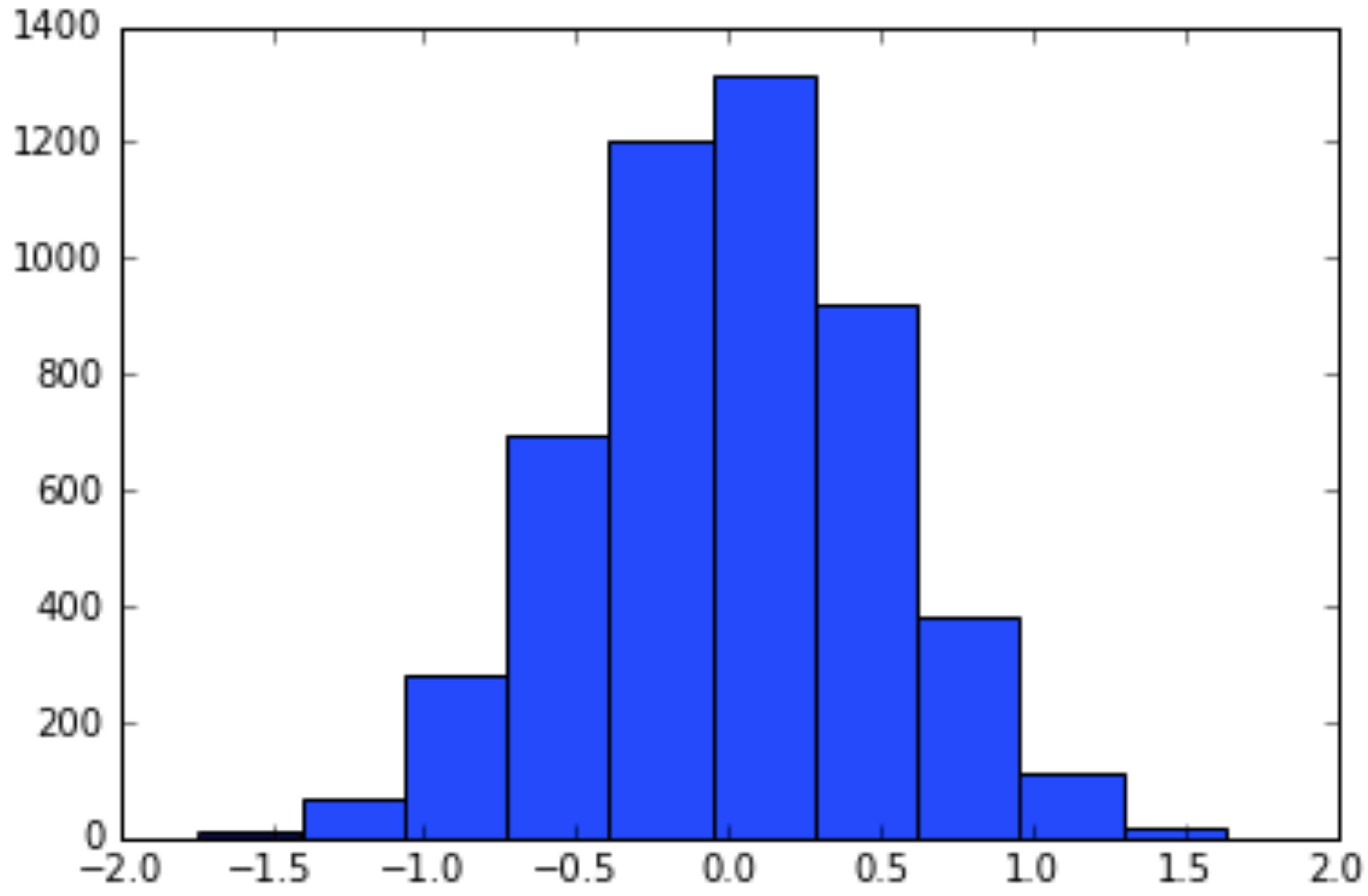


1) generate x,y from U
r=sqry(x^2+y^2+z^2)<=1
2) generate sqrt[r], theta and
z=cos(phi) from U.

Probability distribution functions:

The next basic one: Gaussian





$$P(x) = \frac{e^{\frac{(x-\mu)^2}{\sigma^2}}}{\sqrt{(2\pi)}\sigma}$$

Bayes Theorem

$$\underbrace{P(y \mid x)}_{\text{Posterior}} = \frac{\overbrace{P(x \mid y)}^{\text{Likelihood}}\,\overbrace{P(y)}^{\text{Prior}}}{\underbrace{P(x)}_{\text{Evidence}}}$$

The Likelihood, Prior, Evidence and the posterior are probability distributions.

Bayesian inference

Maximum Likelihood and minimum Chisq method gives us the set of parameter that best fit our data. But does not answer our real question: Given the Data we observe, and a model, what is the probability that a set of parameters describes our data?

$$\underbrace{P\left(Model\left(\vec{w}\right)|Data\right)}_{\text{Posterior}} = \frac{\overbrace{\mathscr{L}\left(Data|Model\left(\vec{w}\right)\right)}^{\text{Likelihood}}\ \overbrace{Pr\left(Model(\vec{w})\right)}^{\text{Prior}}}{\underbrace{E\left(Data\right)}_{\text{Evidence}}}$$

$\vec{w}$: Set of free parameters (weights) for a given model

Goal: Find the set of free parameters (weights) that maximize the posterior for given data and priors.

Bayesian inference

Maximum Likelihood and minimum Chisq method gives us the set of parameter that best fit our data. But does not answer our real question: Given the Data we observe, and a model, what is the probability that a set of parameters describes our data?

$$\underbrace{P\left(\vec{w}\,|\,Data\right)}_{\text{Posterior}} = \frac{\overbrace{\mathscr{L}\left(Data\,|\,\vec{w}\right)}^{\text{Likelihood}}\ \overbrace{Pr\left(\vec{w}\right)}^{\text{Prior}}}{\underbrace{E\left(Data\right)}_{\text{Evidence}}}$$

$\vec{w}$: Set of free parameters (weights) for a given model

Goal: Find the set of free parameters (weights) that maximize the posterior for given data and priors.

Bayesian inference

For parameter estimation, we can skip the computation of the evidence since its a normalization and the maximum of the posterior does not depend on it.

$$\underbrace{P\left(\vec{w} \mid Data\right)}_{\text{Posterior}} \propto \underbrace{\mathscr{L}\left(Data \mid \vec{w}\right)}_{\text{Likelihood}} \underbrace{Pr\left(\vec{w}\right)}_{\text{Prior}}$$

For convenience, sometimes is useful to work with the logarithm of the posterior, so that.

$$\ln P \propto \ln \mathscr{L} + \ln Pr$$

Gaussian Likelihood

- Likelihood: The **probability** , under the assumption of a model/theory, to observe the data as was actually observed.

$$\mathscr{L} \longrightarrow P(\text{Data}, \text{Model})$$

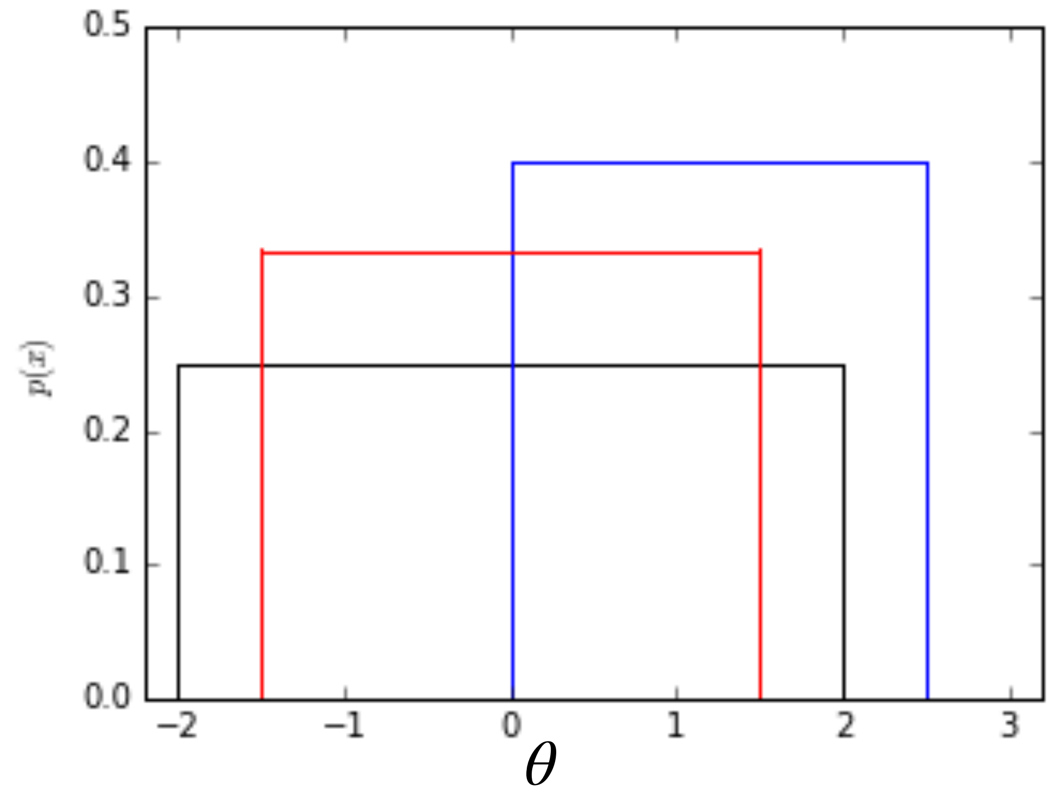
- For data that can be thought as samples of a sequence of normal random variables that have some mean and a variance, the likelihood is Gaussian

$$\mathscr{L} = \Pi_i \frac{1}{2\pi\sigma_i^2} \exp\left(-\frac{\left(y_i - Model\left(x_i, \vec{w}\right)\right)^2}{2\sigma_i^2} \right)$$

$$-\ln \mathscr{L} \propto \sum_i \frac{\left(y_i - Model\left(x_i, \vec{w}\right)\right)^2}{2\sigma_i^2}$$

Prior

Flat Prior



If $\theta \epsilon (\theta_{min}, \theta_{max})$ $\quad Pr(\theta) = \dfrac{1}{\theta_{max} - \theta_{min}}$

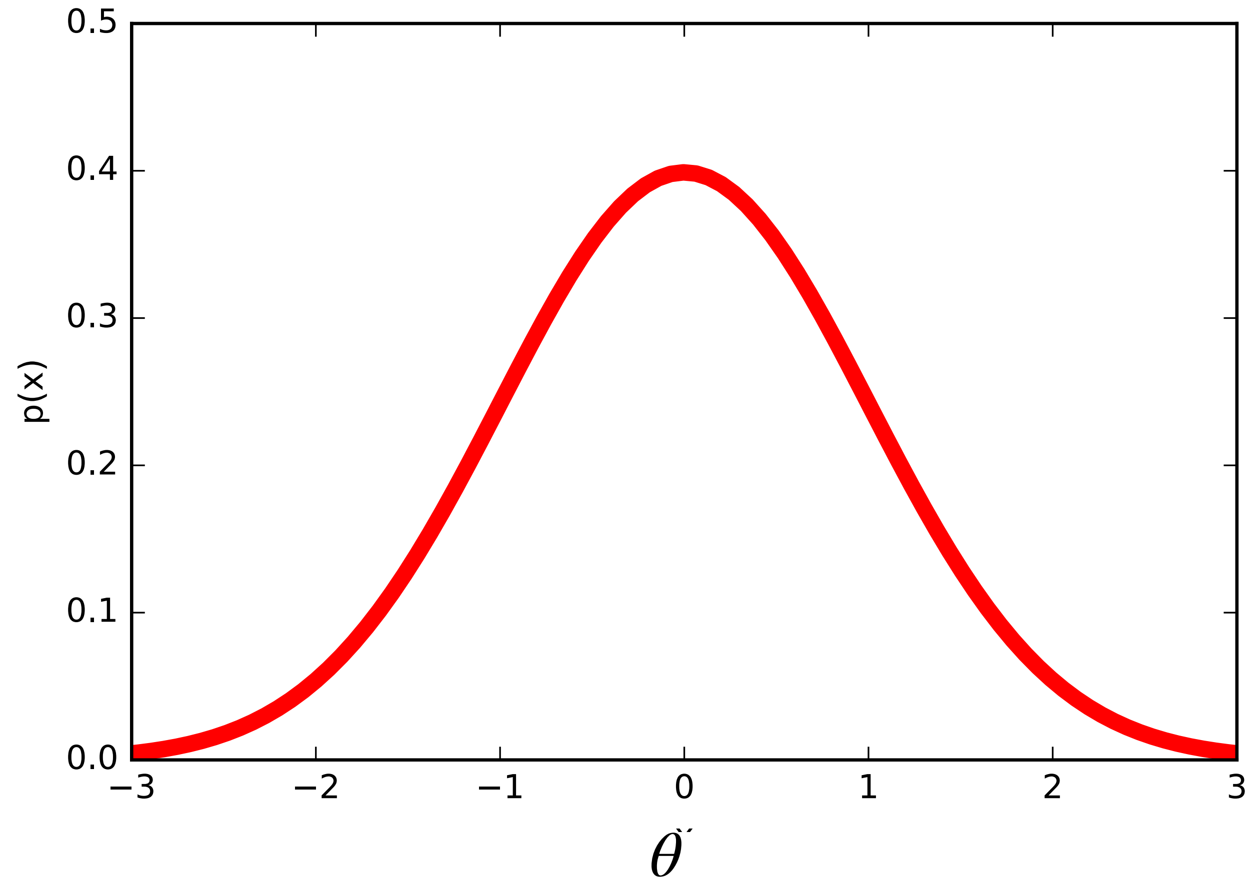
else $\quad Pr(\theta) = 0$

In many cases we ignore the normalization and just set

If $\theta \epsilon (\theta_{min}, \theta_{max})$ $\quad Pr(\theta) = 1$

else $\quad Pr(\theta) = 0$

Gaussian



$$Pr(\theta) \propto e^{(\theta - \mu_\theta)^2 / 2\sigma_\theta^2}$$

Maximum Likelihood/Posterior

How do we maximize the likelihood or the posterior if there 2,3, or more parameters…?

How do we maximize the likelihood or the posterior if we have a complex model?

What if the likelihood or the posterior is not Gaussian…?

MonteCarlo Methods

Technique(s) used to solve a problem based on the generation of random process.

Problems could be:

- Stochastic (probabilistic) in nature. Ej. population growth in terms of survival and reproduction. Virus propagation. Risk on investments...
- Deterministic. Ej. Evaluation of integrals, , solving differential equations, outcome of a controlled experiment, or not so controlled... Complex systems, or with many degrees of freedom (fluids, etc...)

MonteCarlo Methods

Most common applications.

- Optimization
- Numerical integration
- Elements drawn from a particular distribution.

A particular application that summarize the three above is the statistical inference, and most commonly the parameter estimation.

Monte Carlo Markov Chain
Metropolis algorithm

Draw random samples of the free parameters and accept them or reject them according to the posterior (or likelihood) until you reach a maximum of the posterior (likelihood).

1.- Define an initial value of the free parameters.

2.- Draw a new random sample of the free parameters from a gaussian distribution centered at the initial values, and with a predefined dispersion.

3.- Evaluate the (logarithm) posterior of the initial parameters and that of the proposed parameters and compare them.

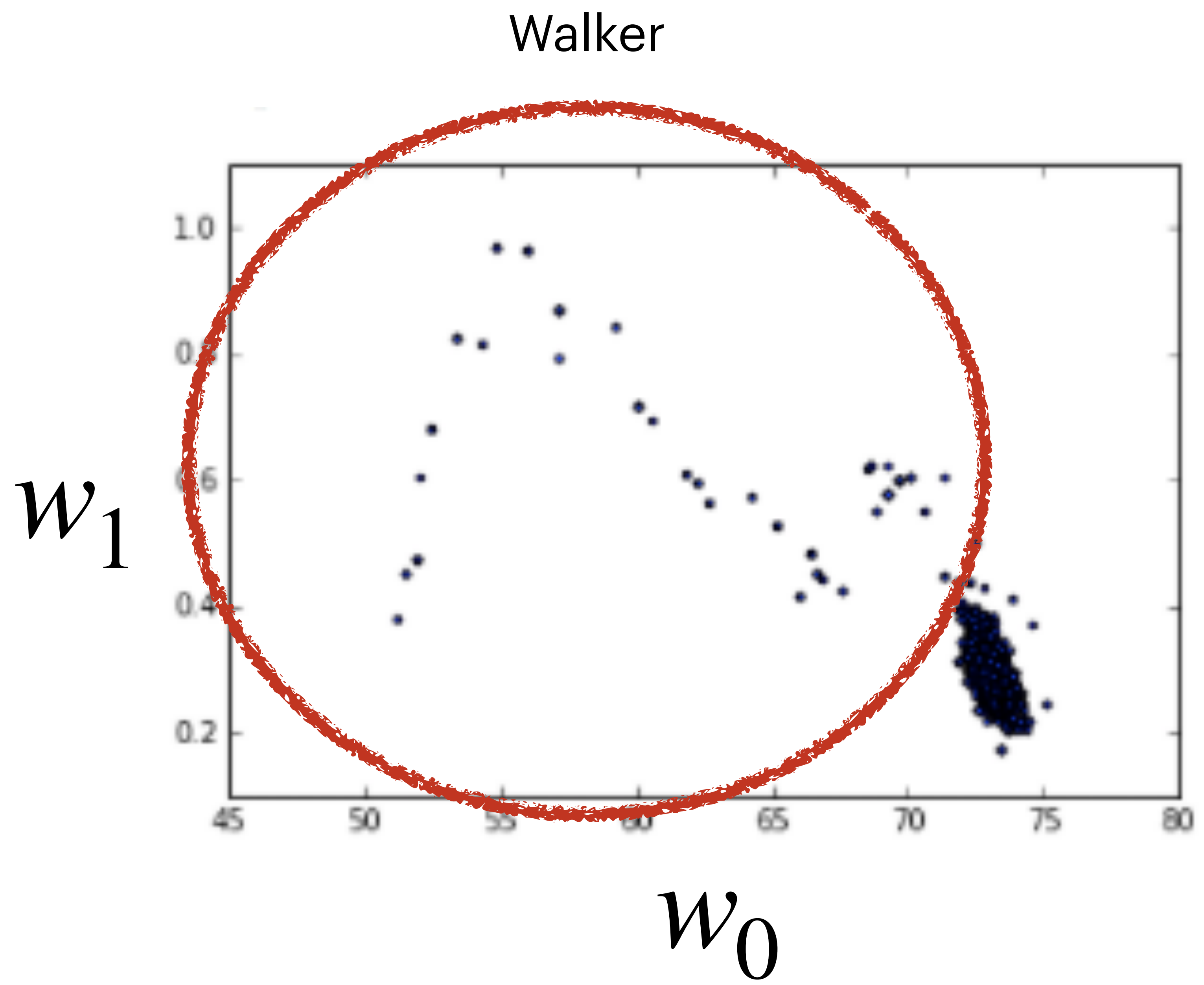
- if the (logarithm) posterior of the proposal is higher than the (logarithm) posterior of the initial parameters accept the proposal and save it.
- if the (logarithm) posterior of the proposal is lower than the (logarithm) posterior of the initial parameters then generate a random number between cero and one, if the ratio of the (logarithm) posterior of the proposal and the initial parameter is larger than such number then you accept the proposal and save it, if not you discard the proposal and keep the initial value.

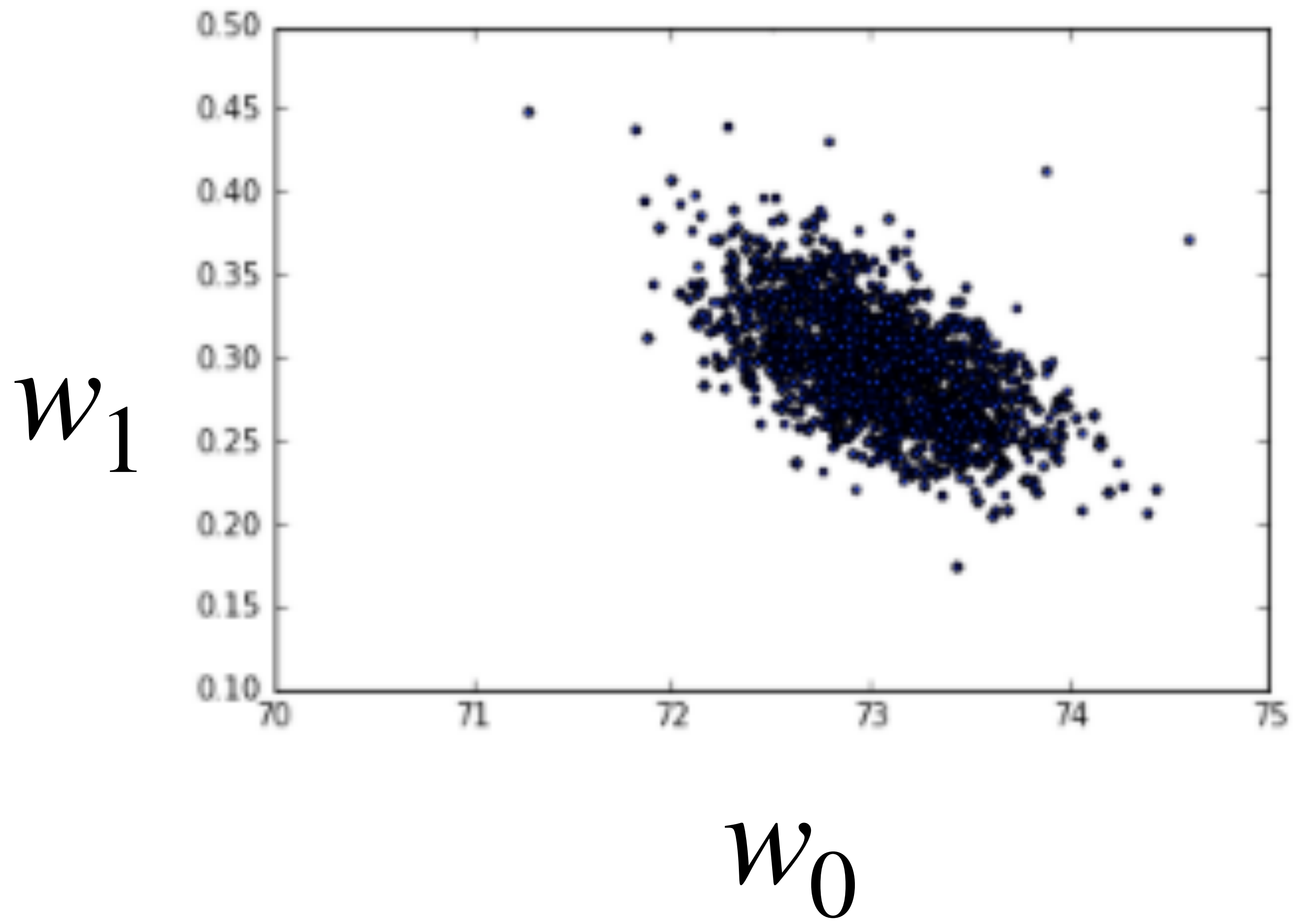
4.- If the proposal was accepted, replace the initial value by such proposal, if not keep the initial values and repeat step

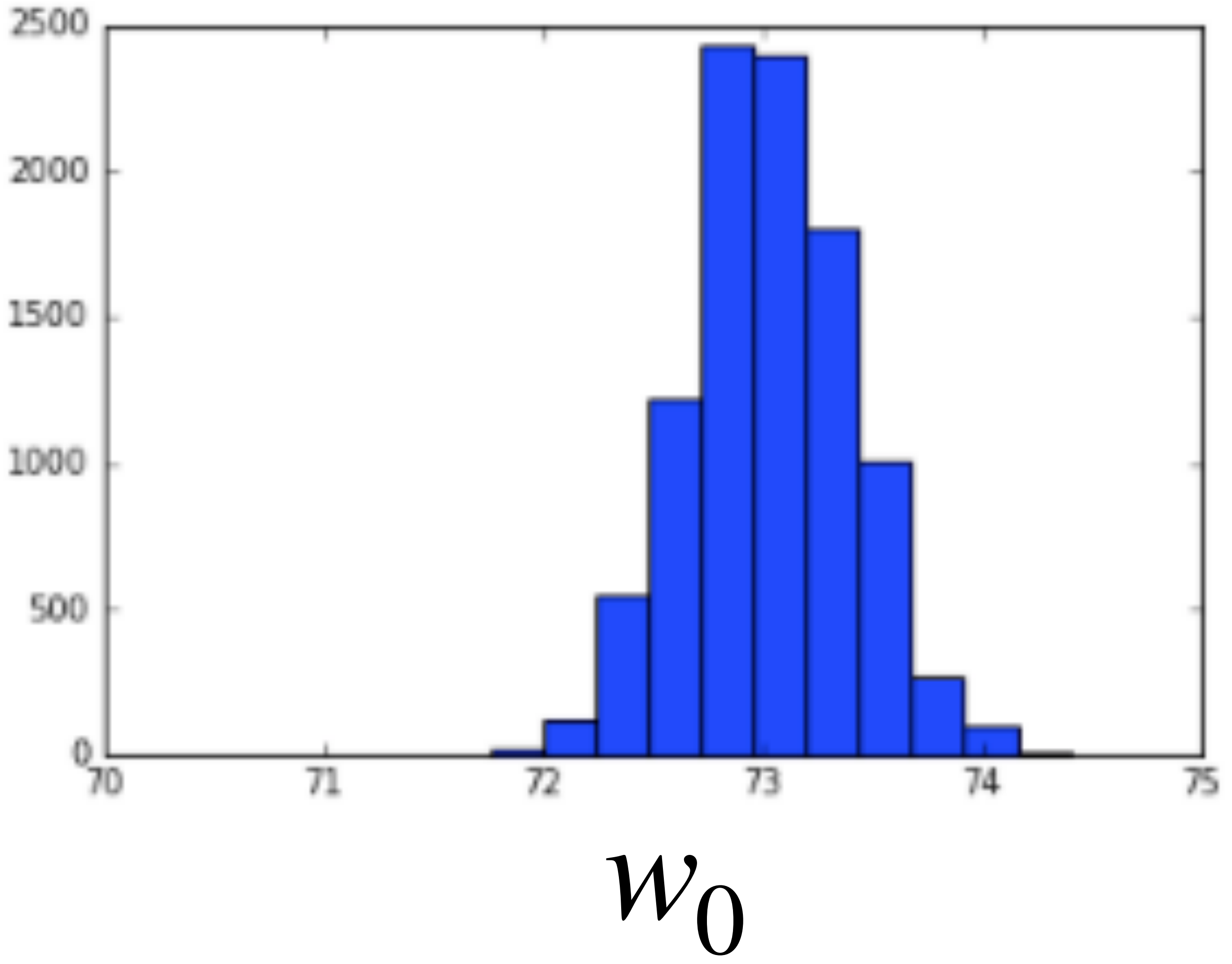
After many steps, look at the resultant distribution (the chains) of parameters, i.e., the likelihood/posterior…

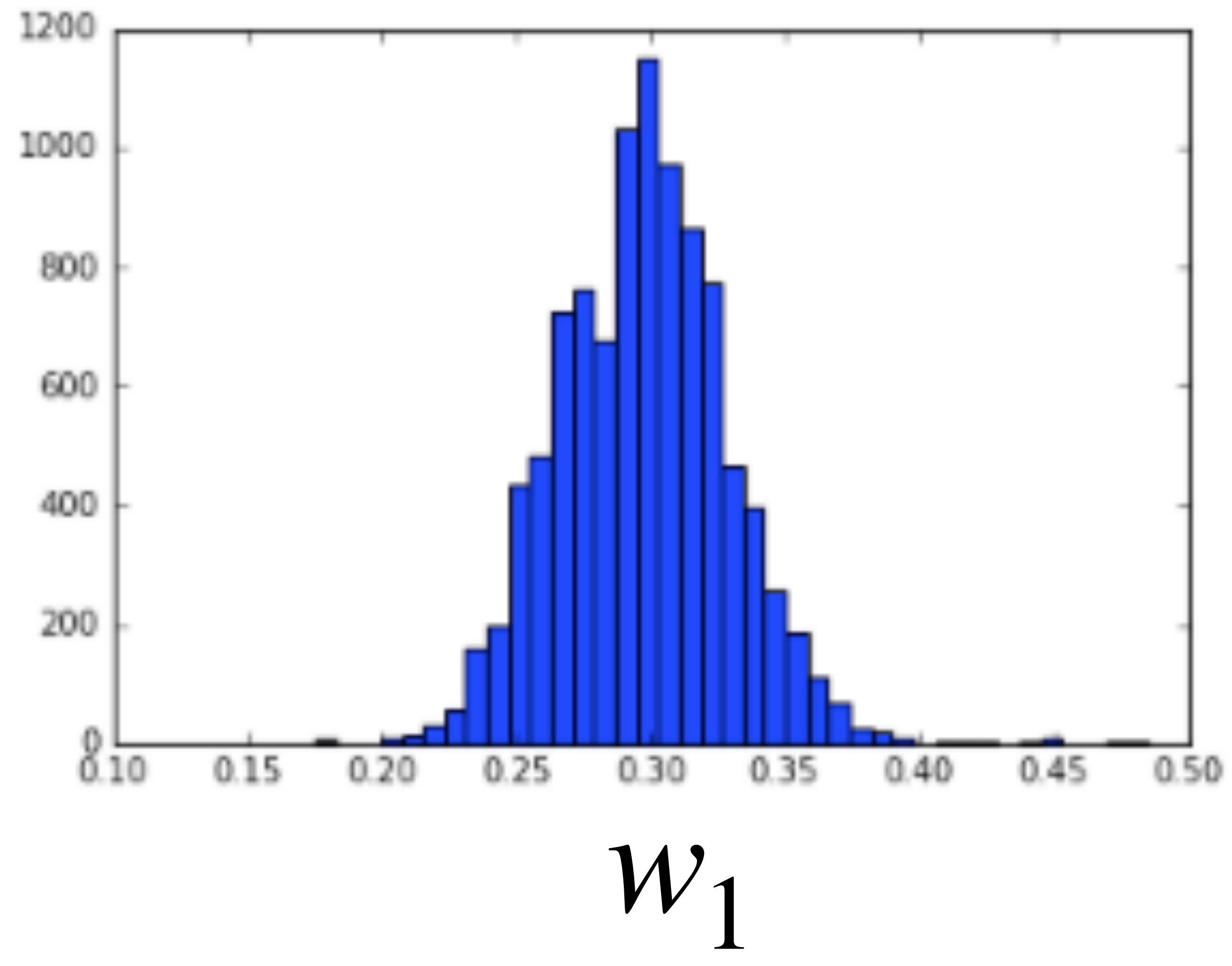
NEXT> Check convergence

Monte Carlo Markov chain









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- MACSS 2016 LEcture notes.